Heart Disease Classification using Repeated Nested Cross-Validation: A Reproducible Pipeline with SHAP-based Interpretation on the Cleveland UCI Dataset

Giorgos Boulogeorgos
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Abstract

We present a repeated nested cross-validation (rnCV) pipeline for binary classification of coronary artery disease on the Cleveland subset of the UCI Heart Disease dataset (242 patients, 13 features). All preprocessing and feature-selection steps are contained within cross-validation folds to prevent data leakage. Seven classifiers were compared under default and tuned hyperparameters across $R = 10$ repetitions of 5×3 stratified rnCV with 50 Optuna trials per inner study. Linear Discriminant Analysis (LDA) was identified as the winning algorithm, achieving median MCC 0.632 (95% CI: 0.623–0.669), AUC 0.895 (0.879–0.911), and PR-AUC 0.894 (0.875–0.916) across 50 outer-test scores. Permutation-importance feature selection in the outer loop identified a stable five-feature set — thallium scan, vessel count, chest pain type, sex, and exercise-induced angina — selected in $\geq 80\%$ of folds and slightly improving median MCC to 0.667. SHAP interpretation confirmed that the model's decision logic aligns with established cardiology knowledge, including the silent-ischaemia pattern in chest pain. An error analysis on a held-out validation split identified the thallium scan as the dominant driver of both correct and incorrect predictions, suggesting a probability-based clinical triage strategy. The pipeline is serialised as a single deployable artefact accepting raw clinical input.

Code Availability: The full implementation and reproducible pipeline are available at: `https://github.com/GiorgosBoulogeorgos/Machine-learning-in-computational-biology-assignment2-`

1 Introduction

Coronary artery disease (CAD) remains the leading cause of mortality worldwide, accounting for approximately 9 million deaths annually [1]. Early and accurate identification of patients at risk is central to clinical management, since timely intervention with lifestyle modification, pharmacotherapy, or revascularisation can substantially alter long-term outcomes. The conventional diagnostic pathway combines clinical history, non-invasive testing (resting and exercise ECG, thallium imaging), and ultimately invasive coronary angiography. While angiography remains the gold standard, its cost, invasiveness, and limited availability motivate the development of decision-support tools that can identify which patients are most likely to benefit from referral.

Machine learning has been applied extensively to this task, with the UCI Heart Disease dataset [2] serving as one of the most widely studied benchmarks in medical machine learning. Despite a sample size of only a few hundred patients, the dataset captures the heterogeneity of routine cardiology workup — demographic, symptomatic, biochemical, and imaging-derived features — and remains relevant for evaluating modern modelling pipelines under realistic small-data constraints. Performance estimates reported on this dataset have ranged widely [2, 3], reflecting differences in preprocessing, validation strategy, and the level of methodological rigour applied to avoid optimistic bias.

The central methodological challenge in clinical machine learning on small datasets is producing a generalisation estimate that is both unbiased and reproducible. Standard k-fold cross-validation is well known to be biased upward when used simultaneously for hyperparameter selection and performance reporting, because the same data informs both decisions [4]. Repeated nested cross-validation (rnCV) [5, 6] addresses this by introducing a strict separation between an outer loop for performance estimation and an inner loop for hyperparameter tuning, with the procedure repeated multiple times to reduce variance. Combined with strict containment of all preprocessing and feature-selection steps inside the cross-validation folds, rnCV provides a defensible workflow for reporting generalisation performance on small datasets.

This work presents a complete rnCV-based machine learning pipeline for binary heart disease classification on the Cleveland subset of the UCI dataset. The pipeline is implemented following object-oriented design and addresses the assignment requirements through five specific contributions:

1. **Exploratory data analysis.** A systematic EDA characterises the dataset's structure, identifies the two missing values, the cholesterol outlier, the mild class imbalance, and the most class-discriminative features. Findings inform every subsequent preprocessing decision.

2. **Object-oriented rnCV implementation.** A reusable `RepeatedNestedCV` class encapsulates the full pipeline, accepting a list of estimators with their hyperparameter spaces. Stratified folds, per-repetition seeding, and strict in-fold preprocessing prevent leakage; non-parametric bootstrap CIs quantify uncertainty around median metrics.

3. **Algorithm comparison and winner selection.** Seven algorithms spanning linear, probabilistic, discriminant, and ensemble paradigms are compared under both default and tuned hyperparameters. Performance is reported on a battery of clinically aligned metrics including MCC, ROC-AUC, PR-AUC, balanced accuracy, F1, recall, specificity, and precision. Linear Discriminant Analysis is identified as the winner.

4. **Stable feature selection.** Permutation-importance feature selection inside the outer loop identifies a five-feature stable subset (`thal`, `ca`, `cp`, `sex`, `exang`) that matches or improves the full-feature performance while substantially reducing the model's input requirements — a result of practical importance for deployment.

5. **Final model and SHAP interpretation.** A complete deployable pipeline is trained on all available data and serialised. Post-hoc SHAP analysis [7] confirms that the model's decision logic aligns with established cardiology knowledge, including the well-documented silent-ischaemia signature in the chest-pain feature.

A bonus error analysis on a held-out validation split additionally identifies the dominant error mechanism (over-reliance on the thallium scan when it is atypical for the patient's true disease status) and produces a clinician-oriented summary suitable for guiding deployment decisions.

The remainder of this report is structured as follows. Section 2 describes the dataset, preprocessing, rnCV pipeline, and evaluation methodology. Section 3 reports EDA findings, algorithm comparison, feature selection, final model performance with SHAP interpretation, and the bonus error analysis. Section 4 summarises the main findings, discusses limitations, and outlines directions for further investigation.

2 Materials and Methods

2.1 Dataset

The Cleveland subset of the UCI Heart Disease dataset [2] was used, comprising 242 patient records and 13 clinical attributes. The binary classification target indicates the presence (`num` = 1) or absence (`num` = 0) of coronary artery disease. The 13 features include demographic descriptors (age, sex), clinical measurements (resting blood pressure `trestbps`, serum cholesterol `chol`, fasting blood sugar `fbs`, maximum heart rate `thalach`, ST depression `oldpeak`), categorical clinical findings (chest pain type `cp`, resting ECG `restecg`, exercise-induced angina `exang`, ST-segment slope `slope`, thallium scan `thal`), and the number of major vessels affected on fluoroscopy (`ca`). Two missing values were present in the dataset — one each in `ca` and `thal` — and one extreme outlier was identified in `chol` (564 mg/dL). The class distribution is mildly imbalanced (54.1% no-disease, 45.9% disease), corresponding to an imbalance ratio of approximately 1.18:1.

2.2 Preprocessing

Preprocessing was implemented as an `sklearn ColumnTransformer` that applies different transformations to feature subsets. Continuous features (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`) were standardised to zero mean and unit variance via `StandardScaler`, with mean imputation applied beforehand as a defensive default. Categorical and binary features were imputed using the most-frequent (mode) value, addressing the two missing values in `ca` and `thal` without introducing distortion to the other categorical features. No scaling was applied to categorical features, as this would corrupt their integer encoding.

The transformer was constructed as a fresh, unfitted instance for each cross-validation fold and incorporated as the first step of a pipeline that ended with the classifier. This design ensures that imputation statistics and scaling parameters are computed exclusively on training data within each fold, with no leakage from validation or test data.

2.3 Repeated Nested Cross-Validation

Generalisation performance was estimated via repeated nested cross-validation (rnCV) using $R = 10$ independent repetitions of $N = 5$-fold outer $\times$ $K = 3$-fold inner stratified cross-validation.

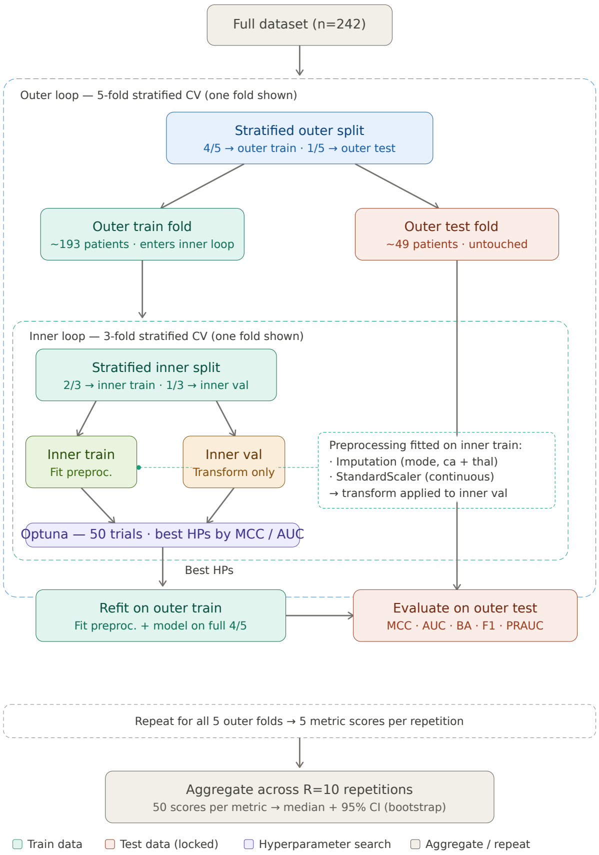


Figure 1. One iteration of the rnCV pipeline. The outer loop estimates generalisation performance; the inner loop performs hyperparameter tuning. Preprocessing and (in Task 4) feature selection are fitted strictly on training folds at each level.

This produces 50 outer-test scores per algorithm per metric. Stratified splitting preserved the class ratio in every fold. To ensure reproducibility, the random seed for repetition r was set to $r_{\text{seed}} = r_{\text{base}} + r$, with $r_{\text{base}} = 42$, propagating identically to fold-splitters, classifier seeds, and the Optuna sampler. Figure 1 illustrates one iteration of the pipeline.

The inner loop performs hyperparameter optimisation via Optuna [8] using the Tree-structured Parzen Estimator (TPE) sampler with 50 trials per inner study. For each candidate hyperparameter configuration, the inner-validation MCC is averaged across the three inner folds before being returned to the optimiser. Aggregating across folds before optimisation favours hyperparameters that generalise across inner-train splits over those that exploit a single favourable split. The Matthews Correlation Coefficient (MCC) was selected as the inner-loop metric for three reasons: (i) it incorporates all four cells of the confusion matrix, (ii) it is invariant under class swap, and (iii) it remains well-defined under mild class imbalance, unlike F1.

The complete metric battery computed on each outer-test fold comprised MCC, ROC-AUC, balanced accuracy, F1, recall, specificity, precision, and PR-AUC. For each algorithm and metric, the median across the 50 outer-test scores was reported as the central estimate, with non-parametric 95% confidence intervals (CIs) constructed by percentile bootstrap resampling (10,000 resamples). Pairs of algorithms with overlapping CIs on median MCC were treated as statistically indistinguishable on this primary metric.

2.4 Algorithm Comparison

Seven classifiers spanning linear, probabilistic, discriminant, and ensemble paradigms were compared: Logistic Regression with Elastic Net regularisation (LR), Gaussian Naive Bayes (GNB), Linear Discriminant Analysis with shrinkage (LDA), Random Forest (RF), LightGBM, XGBoost, and CatBoost. Each algorithm was first evaluated under default hyperparameters using $R = 10$ iterations of 5-fold stratified CV (the inner loop disabled, $K = \emptyset$), establishing a fair baseline against which the gain from tuning could be assessed. The full rnCV pipeline was then run with algorithm-specific hyperparameter spaces summarised in Table 1. Ranges for regularisation parameters were defined on a log scale to reflect their multiplicative nature; integer ranges (tree depth, number of estimators) covered values that are conventional for small tabular datasets.

The winning algorithm was selected by ranking on median MCC. Ties (overlapping 95% CIs) were to be broken on median AUC and PR-AUC.

Table 1. Hyperparameter search spaces per algorithm.

Algorithm	**Hyperparameter**	**Range**
LR	C	log $[10^{-3}, 10^{1}]$
	l1_ratio	$[0, 1]$
GNB	var_smoothing	log $[10^{-12}, 10^{-6}]$
LDA	solver	{lsqr, eigen}
	shrinkage	$[0, 1]$
RF	n_estimators	$\{100, \ldots, 500\}$ step 50
	max_depth	$\{3, \ldots, 20\}$
	min_samples_split	$\{2, \ldots, 20\}$
	min_samples_leaf	$\{1, \ldots, 10\}$
	max_features	{sqrt, log2}
LightGBM	n_estimators	$\{100, \ldots, 500\}$ step 50
	learning_rate	log $[10^{-3}, 0.3]$
	num_leaves	$\{8, \ldots, 64\}$
	max_depth	$\{3, \ldots, 12\}$
	min_child_samples	$\{5, \ldots, 30\}$
	reg_alpha, reg_lambda	log $[10^{-3}, 1]$
XGBoost	n_estimators	$\{100, \ldots, 500\}$ step 50
	learning_rate	log $[10^{-3}, 0.3]$
	max_depth	$\{3, \ldots, 10\}$
	min_child_weight	$\{1, \ldots, 10\}$
	subsample, colsample_bytree	$[0.6, 1.0]$
	reg_alpha, reg_lambda	log $[10^{-3}, 1]$
CatBoost	iterations	$\{100, \ldots, 500\}$ step 50
	learning_rate	log $[10^{-3}, 0.3]$
	depth	$\{3, \ldots, 10\}$
	l2_leaf_reg	$[1, 10]$

2.5 Feature Selection

Following identification of the winning algorithm, model-agnostic feature selection was performed using permutation importance [9, 10] with MCC as the scoring metric. The selector was applied within the outer loop of the rnCV: for each outer training fold, a default-hyperparameter LDA was fitted, permutation importances were estimated with 10 repeats per feature, and the top-k features were retained. The same feature subset was then propagated to the inner loop and final outer-test evaluation. The locked outer test fold was never seen by the selector at any stage.

The number of retained features k was determined by sweeping $k \in \{3, 4, \ldots, 12\}$ and selecting the value maximising median MCC across the 50 outer-test scores. This data-driven optimum was preferred over a more parsimonious criterion (e.g., the smallest k whose 95% CI overlaps the full-feature CI) because it captures the most predictive feature set the data supports without overparametrisation. To assess feature stability, the selection frequency of each feature was recorded across all 50 fold $\times$ repetition outcomes at the chosen k. Features selected in $\geq 80\%$ of folds, in line with conventions from the stability selection literature [11], were retained as the *stable feature set* for downstream deployment.

2.6 Final Model and Persistence

The final model was trained on the entire dataset using the stable feature set. Optimal hyperparameters were determined by a single (non-nested) 5-fold stratified CV over the LDA hyperparameter space using Optuna with 50 trials. The deployable pipeline was constructed in three stages: (i) a custom `FeatureSubsetSelector` accepting raw 13-column input and projecting onto the stable feature subset, (ii) the `ColumnTransformer` for imputation and standardisation, and (iii) the LDA classifier with the chosen hyperparameters. The complete pipeline was fitted on all 242 patients and serialised using `pickle` as a single artefact (`models/final_model.pkl`). To verify deployability, the pickled artefact was reloaded in a fresh state and verified to produce predictions on raw 13-column input without external preprocessing.

2.7 Model Interpretation with SHAP

SHAP (SHapley Additive exPlanations) values [7] were computed on the final model fitted on all available data, providing model interpretation grounded in cooperative game theory. The model-agnostic `KernelExplainer` was selected, as LDA is neither tree-based (precluding `TreeExplainer`) nor straightforwardly compatible with SHAP's `LinearExplainer` due to its probabilistic output structure. To keep computation tractable, the background distribution was summarised using k-means clustering with 50 centroids; this approximation preserves the input distribution while reducing the cost of the marginal-expectation estimates.

SHAP values were computed for each of the 242 patients on the stable feature subset, yielding (i) a global ranking of mean absolute SHAP per feature, (ii) a beeswarm summary plot showing both magnitude and direction of effect, and (iii) waterfall plots decomposing individual predictions into per-feature contributions for two illustrative cases (one correctly predicted disease, one correctly predicted healthy).

2.8 Bonus Task: Error Analysis

A complementary error analysis was performed using a single 70/30 stratified train/validation split (`random_state` $= 42$), with the model architecture and hyperparameters fixed to those of the final model. After predicting on the validation set, each patient was assigned to one of four categories: true positive (TP), true negative (TN), false positive (FP), or false negative (FN). Demographic and clinical features were compared across these categories using Mann-Whitney U tests for continuous variables and Fisher's exact tests for categorical variables. SHAP values were computed on the validation set, and per-feature SHAP distributions were compared across the four categories to identify features whose contribution patterns differ between correct and incorrect predictions. Subgroup-specific error rates were computed by stratifying on sex, age band, and key clinical features.

2.9 Implementation and Reproducibility

The pipeline was implemented in Python 3.10 using `scikit-learn` (1.3), `optuna` (3.4), `shap` (0.44), `lightgbm`, `xgboost`, and `catboost`. The code follows object-oriented design, with the rnCV procedure encapsulated in a single `RepeatedNestedCV` class (in `src/rncv.py`) that accepts a list of `EstimatorConfig` objects pairing each estimator with its hyperparameter-space callable. Random seeds were fixed throughout. `Optuna` was set to single-thread execution to avoid stochasticity in the TPE sampler; per-tree parallelism (`n_jobs` $= -1$) was retained only for Random Forest, where each tree is independently seeded. The full repository is available at: GitHub Repository.

3 Results and Discussion

3.1 Exploratory Data Analysis

The dataset was found to be largely clean: 242 patients with no duplicate rows and only two missing entries (one each in `ca` and `thal`). The mild class imbalance (54.1% no-disease, 45.9% disease, Fig. 2) confirmed that performance metrics insensitive to imbalance — particularly MCC, ROC-AUC, balanced accuracy, and PR-AUC — should be prioritised over raw accuracy.

Continuous features showed marked between-class differences for `thalach` and `oldpeak` (both Mann-Whitney $p < 0.001$), modest separation

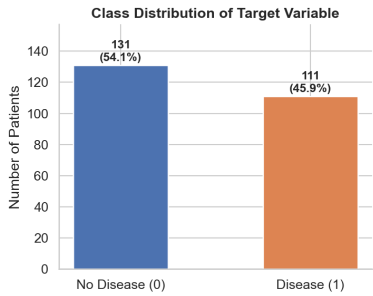


Figure 2. Class distribution of the target variable. The 1.18:1 ratio motivates the use of stratified folds and class-imbalance-robust metrics throughout the pipeline.

for `age` ($p < 0.001$), and weak separation for `trestbps` and `chol` (Fig. 3). Patients with disease achieved substantially lower maximum exercise heart rates (median ∼140 vs ∼160 bpm) and higher ST-depression values, consistent with reduced cardiac reserve and exercise-induced ischaemia.

The categorical-feature distributions (Fig. 4) revealed several findings of clinical interest. Asymptomatic chest pain (`cp` = 4) was paradoxically the chest-pain category most strongly associated with disease — a well-established but counter-intuitive pattern attributable to silent ischaemia and the selection bias of a clinical referral cohort. Reversible thallium defects (`thal` = 7) were strongly associated with disease, as was a higher number of major vessels affected (`ca`). Exercise-induced angina (`exang`) showed a clear discriminative pattern, with ∼76% of patients reporting it having disease versus ∼31% of those without it. Sex showed an epidemiologically expected pattern with higher disease rates in males.

The Pearson correlation matrix (Fig. 5) confirmed that no pair of features exhibited high collinearity ($|r| < 0.6$ throughout), and the strongest target correlations were observed for `thal`, `ca`, `exang`, `thalach`, `oldpeak`, and `cp`. Principal component analysis (Fig. 6) showed that the first two principal components explained only 37% of total variance, with partial class separation visible along PC1. The remaining variance was distributed broadly across components, suggesting that aggressive linear dimensionality reduction would discard meaningful signal — non-linear classifiers, or linear classifiers operating on the full 13-dimensional feature space, were therefore more appropriate than 2D PCA-based modelling.

These EDA findings motivated several preprocessing decisions: mode imputation for the categorical features with missing values, `StandardScaler` for continuous features only, retention of the `chol` = 564 mg/dL outlier as a clinically plausible extreme, and stratified cross-validation throughout. The set of features identified as most informative — `cp`, `thal`, `ca`, `thalach`, `oldpeak`, `exang` — would later be revisited in the feature-selection (Section 3.3) and SHAP (Section 3.4) analyses, providing a useful triangulation across the EDA, model-driven feature selection, and post-hoc interpretation.

3.2 Algorithm Comparison

Table 2 reports per-algorithm performance under default hyperparameters, and Table 3 reports the tuned-rnCV results. Both are summarised as median with 95% bootstrap CI across the 50 outer-test scores.

Several patterns emerge. The default-hyperparameter baseline (Table 2) places LDA at the top with median MCC 0.667, ahead of GaussianNB (0.642) and LogisticRegression (0.640). The tree-based ensembles — RF, LightGBM, XGBoost, and CatBoost — trail behind, with median MCC values of 0.553–0.590. After hyperparameter tuning (Table 3), the gradient-boosting methods improve substantially in MCC (XGBoost: 0.569 → 0.628; LightGBM: 0.553 → 0.582; CatBoost: 0.588 → 0.623), while the linear and discriminant methods change very little. This pattern is illustrated graphically in Figure 7. The asymmetry is intuitive: tree-based methods have larger and more consequential hyperparameter spaces than methods like LDA or GNB, so they benefit more from tuning. The convergence after tuning means the ranking is much closer at the top.

The forest plot of tuned median MCC with 95% CIs (Fig. 8) shows GaussianNB highest at 0.642, followed by LDA (0.632), XGBoost (0.628), RandomForest (0.627), LogisticRegression (0.627), CatBoost (0.623), and LightGBM (0.582). The CIs of all algorithms except LightGBM overlap one an-

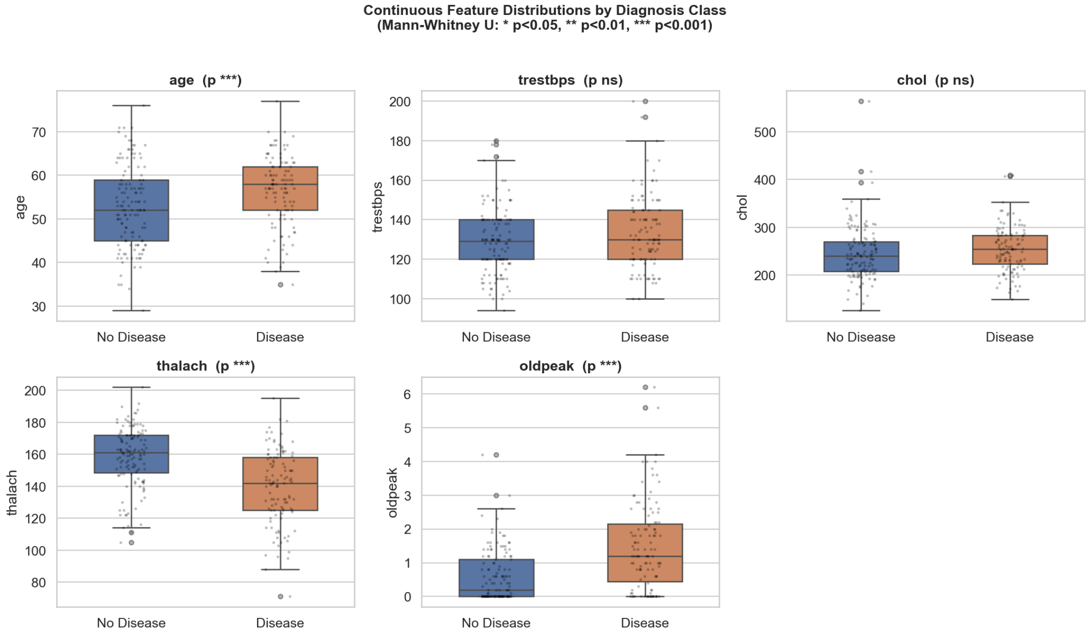


Figure 3. Continuous-feature distributions stratified by diagnosis class. `thalach` and `oldpeak` show the strongest separation, both consistent with established markers of coronary ischaemia. Annotated p-values are from Mann-Whitney U tests.

Table 2. Default-hyperparameter baseline. $R = 10$ iterations of 5-fold stratified CV, no inner loop. Values: median [95% bootstrap CI].

Algorithm	MCC	AUC	PR-AUC	Balanced Acc
LogisticRegression	0.640 [0.590, 0.678]	0.892 [0.876, 0.906]	0.893 [0.866, 0.912]	0.815 [0.794, 0.833]
GaussianNB	0.642 [0.607, 0.687]	0.892 [0.878, 0.907]	0.886 [0.860, 0.901]	0.817 [0.798, 0.842]
LDA	**0.667 [0.628, 0.707]**	**0.897 [0.881, 0.911]**	**0.897 [0.874, 0.911]**	**0.823 [0.809, 0.849]**
RandomForest	0.590 [0.577, 0.624]	0.878 [0.864, 0.901]	0.876 [0.861, 0.896]	0.794 [0.779, 0.809]
LightGBM	0.553 [0.538, 0.590]	0.856 [0.837, 0.866]	0.860 [0.833, 0.871]	0.775 [0.766, 0.793]
XGBoost	0.569 [0.523, 0.593]	0.859 [0.837, 0.876]	0.860 [0.844, 0.871]	0.779 [0.759, 0.796]
CatBoost	0.588 [0.550, 0.626]	0.877 [0.865, 0.892]	0.875 [0.865, 0.906]	0.787 [0.774, 0.809]

other (Fig. 9), indicating that on this 242-patient dataset, the top six algorithms are statistically indistinguishable on median MCC alone. LightGBM is the only algorithm whose CI does not overlap LDA's or GaussianNB's, marking it as the only clearly inferior choice.

Tie-breaking by median AUC and PR-AUC (Fig. 10) refines the picture. LDA achieves the highest median AUC at 0.895 and the highest PR-AUC at 0.894, narrowly ahead of GaussianNB (AUC 0.892, PR-AUC 0.886). LDA also achieves the highest specificity (0.885), tied with LogisticRegression and RandomForest. Combining the MCC, AUC, PR-AUC, and specificity rankings, **LDA was selected as the winning algorithm**. The choice is also defensible on simplicity grounds: among the algorithms statistically tied at the top, LDA has the smallest hyperparameter space and is the most interpretable.

This finding is consistent with the broader literature on small clinical tabular datasets: when n is on the order of a few hundred and the feature count is modest, simple linear and discriminant methods routinely match or exceed the performance of gradient-boosting ensembles. LDA's strong distributional assumptions (Gaussian class-conditionals, shared covariance) act as a powerful regulariser that is helpful when the data are insufficient to constrain a more flexible model. The result is also a useful counter-example to the assumption that gradient-boosting is universally optimal for tabular data.

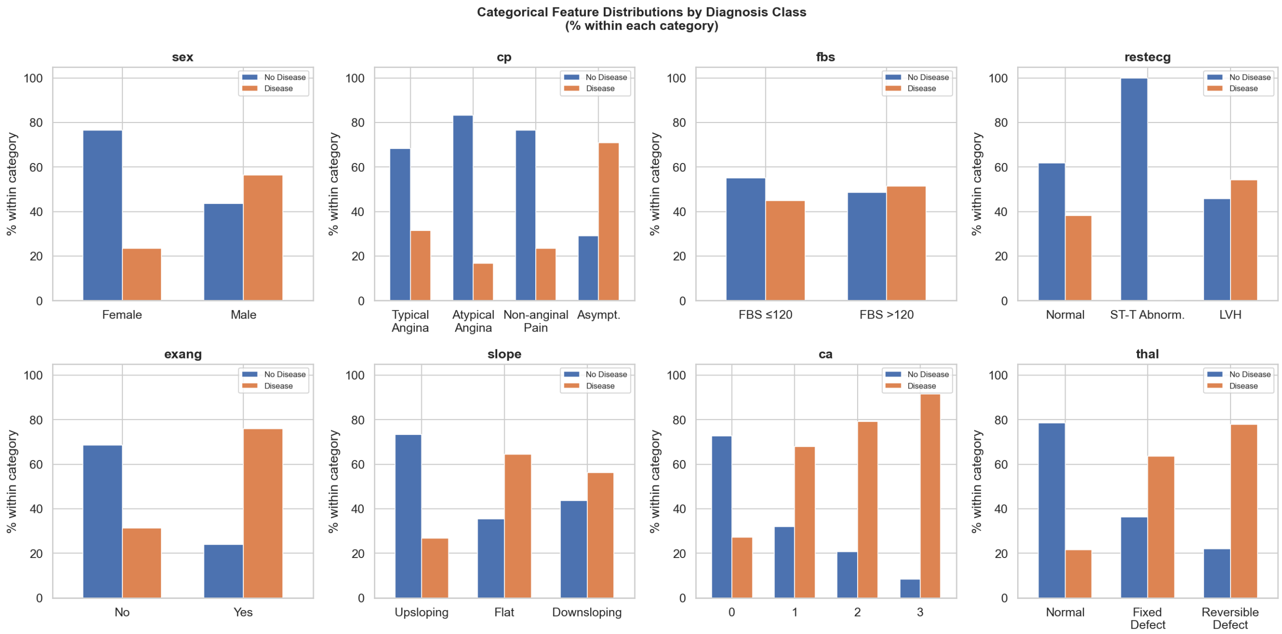


Figure 4. Categorical-feature distributions stratified by diagnosis class. The `cp`, `thal`, `ca`, and `exang` panels each show clear class-discriminative patterns. The `cp` panel illustrates the silent-ischaemia phenomenon: asymptomatic chest pain (rightmost bar group) is the category most strongly enriched for disease.

Table 3. Tuned-rnCV results. $R = 10$ iterations of 5-fold outer $\times$ 3-fold inner CV, 50 Optuna trials per inner study. Values: median [95% bootstrap CI].

Algorithm	**MCC**	**AUC**	**PR-AUC**	**Balanced Acc**
LogisticRegression	0.627 [0.594, 0.667]	0.891 [0.872, 0.907]	0.896 [0.864, 0.912]	0.811 [0.790, 0.827]
GaussianNB	**0.642 [0.607, 0.687]**	0.892 [0.878, 0.907]	0.886 [0.860, 0.901]	0.817 [0.798, 0.842]
LDA	0.632 [0.623, 0.669]	**0.895 [0.879, 0.911]**	**0.894 [0.875, 0.916]**	0.815 [0.806, 0.825]
RandomForest	0.627 [0.585, 0.673]	0.887 [0.878, 0.907]	0.893 [0.873, 0.904]	0.809 [0.793, 0.825]
LightGBM	0.582 [0.524, 0.601]	0.868 [0.849, 0.888]	0.869 [0.854, 0.881]	0.783 [0.758, 0.799]
XGBoost	0.628 [0.558, 0.664]	0.888 [0.860, 0.907]	0.880 [0.867, 0.903]	0.799 [0.778, 0.829]
CatBoost	0.623 [0.558, 0.670]	0.892 [0.868, 0.907]	0.888 [0.871, 0.915]	0.808 [0.778, 0.831]

3.3 Feature Selection

The motivating permutation-importance plot (Fig. 11), computed on a single illustrative 80/20 split, ranks the features as `thal`, `ca`, `cp`, `sex`, `trestbps`, `slope`, `exang`, `chol`, `thalach`, `fbs`, `restecg`, `oldpeak`, `age`. The first three features dominate, separated by a clear gap from the remainder.

The performance-vs-k sweep (Fig. 12) shows that median MCC rises sharply from $k = 3$ ($\sim$0.58) to $k = 6$ ($\sim$0.65), peaks in the $k = 7$–9 range (median MCC $\sim$0.667 at $k = 8$), and then slightly declines for $k \geq 10$. The corresponding AUC curve peaks at $k = 9$. This pattern is the textbook bias-variance trade-off: too few features starve the model of signal, while too many introduce noise and increase the effective dimensionality of LDA's covariance estimate beyond what the 242-patient dataset can support. The optimal feature count was therefore set at $k = 8$, achieving median MCC 0.667 and outperforming the full-feature baseline (median MCC 0.632) by $\Delta = +0.035$.

The per-feature selection frequency at $k = 8$ (Fig. 13) revealed five features selected in $\geq 80\%$ of the 50 outer folds: `ca` (100%), `thal` (100%), `sex` (96%), `cp` (96%), and `exang` (82%). The remaining three slots rotated between several borderline features (`thalach` 58%, `fbs` 56%, `slope` 56%, `age` 44%, `trestbps` 34%, `restecg` 34%, `chol` 28%, `oldpeak` 16%), none of which dominated. The selection-frequency distribution is bimodal — a clear gap separates the five stable features from the remaining candidates — so the result is robust to moderate variations in the 80% stability

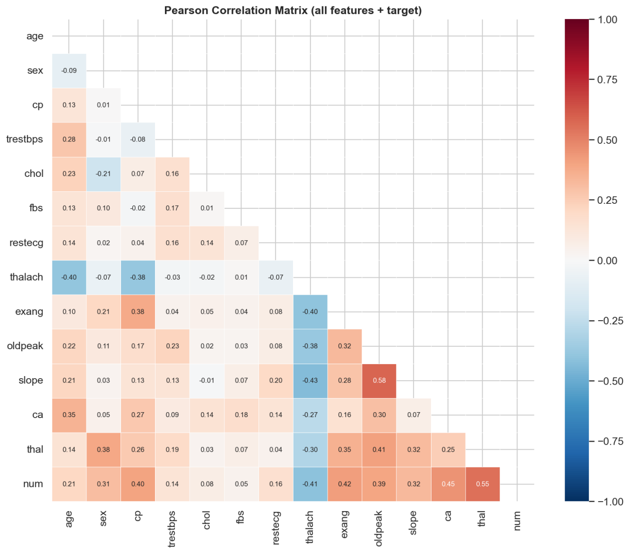


Figure 5. Pearson correlation matrix. No severe collinearity is observed; the bottom row identifies `thal` (0.55), `ca` (0.45), and `exang` (0.42) as the features most linearly correlated with the target.

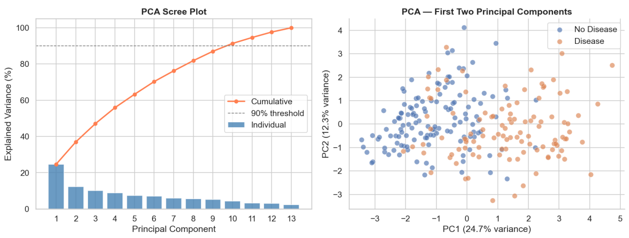


Figure 6. PCA scree plot (left) and 2D projection on the first two principal components (right). The classes are partially separable along PC1, but the 2D projection retains only 37% of the variance.

threshold.

Two design choices warrant explicit comment. First, the optimal feature *count* ($k = 8$) is larger than the stable feature *set* (5 features). The deployed model uses the 5 stable features rather than the 8 selected per fold, prioritising reproducibility and clinical interpretability over a marginal performance gain (the reduced-feature performance comparison in Fig. 14 confirms this is acceptable). Second, the stable feature set comprises only categorical features (`sex`, `cp`, `exang`, `thal`, `ca`); none of the continuous features — `age`, `trestbps`, `chol`, `thalach`, `oldpeak` — entered the stable set despite their visible class-discriminative patterns in the EDA. This is a meaningful clinical finding: the categorical findings of cardiology workup (chest pain type, thallium scan, vessel count, exercise angina, sex) carry sufficient information for LDA to classify reliably without the continuous measurements.

The features selected stably converge with the EDA findings (Section 3.1) on `cp`, `thal`, `ca`, and `exang` as the most informative variables, with `sex` also entering the stable set. The agreement between data-driven feature selection and prior clinical knowledge supports the view that the model is exploiting genuine clinical signal rather than dataset artefacts.

3.4 Final Model and SHAP Interpretation

A single 5-fold CV with Optuna (50 trials) on the entire dataset using the 5 stable features selected the final hyperparameters `solver` = `eigen` and `shrinkage` = 0.006, achieving a 5-fold mean MCC of 0.680. The chosen `shrinkage` is essentially zero — a meaningful finding in itself: with 5 features and 242 patients, the data are sufficient to estimate the LDA covariance reliably without regularisation. If `shrinkage` had landed near 1.0 it would have signalled insufficient samples per parameter. The 5-fold mean MCC (0.680) is slightly higher than the rnCV estimate (0.632) because the latter uses smaller training folds; the rnCV value remains the appropriate honest estimate of generalisation performance.

The complete deployment pipeline (`FeatureSubsetSelector` → `ColumnTransformer` → LDA) was fitted on all 242 patients and serialised to `models/final_model.pkl`. A reload-and-predict sanity check on raw 13-column input confirmed the artefact functions correctly in a fresh state, addressing the deployability requirement of the assignment.

The SHAP global feature-importance ranking (Fig. 15) places `thal` clearly first (mean $|\text{SHAP}| \approx 0.19$), followed by `ca` ($\approx 0.12$), `exang` ($\approx 0.08$), `cp` ($\approx 0.08$), and `sex` ($\approx 0.06$). The fact that `thal` dominates by a factor of $\sim$1.6 over the next feature reflects its strong direct relationship with disease status in this dataset and is consistent with both the EDA correlations and the stability-selection results.

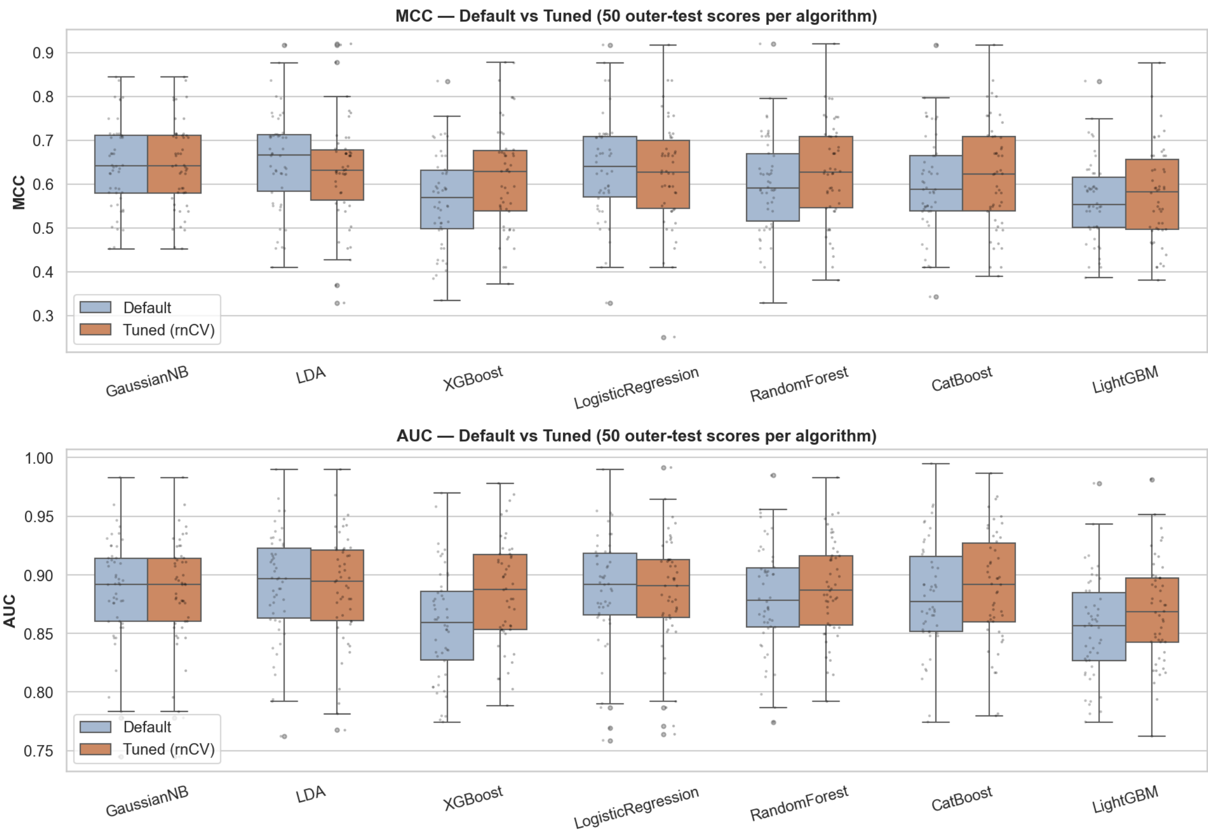


Figure 7. Per-algorithm MCC and AUC distributions across 50 outer-test scores, comparing default and tuned hyperparameters. Tree-based methods benefit visibly from tuning; linear and discriminant methods do not.

Table 4. LDA — full features ($k = 13$) versus reduced ($k = 8$). Median [95% bootstrap CI].

Condition	MCC	AUC	PR-AUC	Balanced Acc
Full ($k = 13$)	0.632 [0.623, 0.669]	0.895 [0.879, 0.911]	0.894 [0.875, 0.916]	0.815 [0.806, 0.825]
Reduced ($k = 8$)	0.667 [0.623, 0.709]	0.899 [0.881, 0.909]	0.901 [0.871, 0.914]	0.827 [0.808, 0.844]

The SHAP summary (beeswarm) plot (Fig. 16) reveals the directional structure of each feature's effect:

- `thal`: bimodal effect. Low values (normal scan, code 3) cluster around SHAP -0.18; high values (reversible defect, code 7) cluster around $+0.20$. This is the strongest and most clinically intuitive pattern.
- `ca`: gradient effect. Zero affected vessels push toward no-disease (SHAP ≈ -0.12); higher vessel counts push progressively toward disease, with some patients reaching SHAP $\approx +0.5$ — the largest single-feature effect in the dataset.
- `exang`: binary effect, as expected. Presence of exercise-induced angina pushes positively (SHAP $\approx +0.10$); absence pushes weakly negative.
- `cp`: gradient effect with the silent-ischaemia signature. Low `cp` codes (typical/atypical angina, codes 1–2) push toward no-disease; high codes (asymptomatic, code 4) push toward disease.
- `sex`: binary effect. Females (`sex` $= 0$) push slightly toward no-disease; males (`sex` $= 1$) push slightly toward disease.

Two individual-prediction explanations are shown in Figure 17. For a correctly predicted disease patient with $P(\text{disease}) = 0.998$, all five features contribute positively, with `ca` $= 3$ (three affected vessels) contributing $+0.26$, `thal` $= 7$ (reversible defect) contributing $+0.14$, and `exang` $= 1$

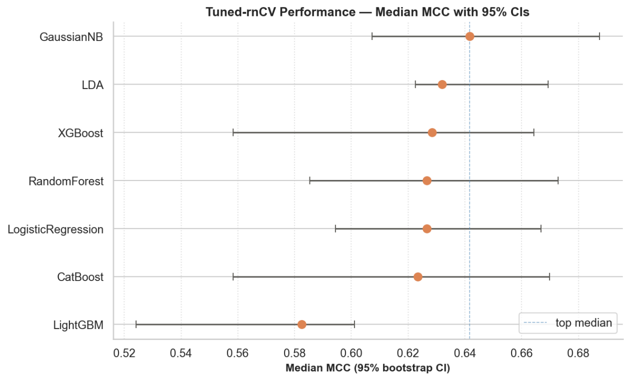


Figure 8. Forest plot of tuned-rnCV median MCC with 95% bootstrap CIs. Six of the seven algorithms have overlapping CIs and are therefore statistically indistinguishable on this metric. LightGBM is the sole outlier.

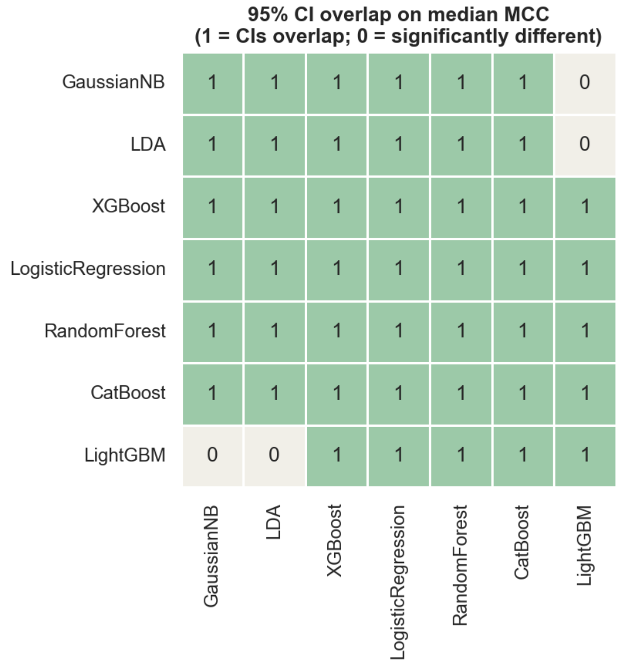


Figure 9. Pairwise 95% CI overlap matrix on median MCC. A value of 1 indicates the two algorithms cannot be distinguished at this confidence level. LightGBM is the only algorithm whose CI does not overlap with both GaussianNB and LDA.

contributing +0.08. For a correctly predicted healthy patient with $P(\text{disease}) = 0.007$, all five features contribute negatively, dominated by `thal` = 3 (normal scan) and `cp` = 1 (typical angina), each contributing −0.12. These waterfall decompositions are the kind of explanation that would support a clinical user in understanding individual predictions, addressing the regulatory transparency expected of any deployed clinical decision-support system.

The cross-task convergence is the clearest result. The same five features — `cp`, `thal`, `ca`, `exang`, `sex` — are repeatedly identified as most important by three independent methods: EDA-based statistical analysis (Section 3.1), permutation-importance feature selection in rnCV outer folds (Section 3.3), and post-hoc SHAP explanations on the final model (this section). This three-way convergence between simple statistical association, model-driven selection, and post-hoc interpretation provides strong evidence that the model has learned genuine clinical signal rather than spurious sample-specific correlations.

3.5 Bonus Task: Error Analysis

The model trained on the 70% partition (169 patients) and evaluated on the 30% validation partition (73 patients) achieved 86.3% accuracy — 36 true negatives, 27 true positives, 6 false negatives, and 4 false positives (Fig. 18). The validation MCC, AUC, and balanced accuracy were 0.723, 0.920, and 0.859 respectively, all comparable to the rnCV estimates, confirming that the chosen architecture generalises consistently.

The ratio of false negatives to false positives (6:4) is clinically relevant: in coronary disease screening, missing disease is more harmful than over-flagging it. This is one motivation for considering a decision threshold below the default 0.5 in deployment, accepting some additional FPs to reduce FNs.

The age comparison (Fig. 19) shows that misclassified patients cluster in the 50-60 age band, with no errors observed below age 50 or above 60 in this validation set. The Mann-Whitney comparison of age between correctly and incorrectly classified patients was not statistically significant ($p > 0.05$), but the visual pattern is striking enough to warrant further investigation in larger samples. The effect is more visible in the sex × age heatmap (Fig. 20): the 50-60 age band exhibits 23.3% error rate among males and 20.0% among females, while the under-50 and over-60 bands exhibit close to zero errors.

The most informative finding emerges from the SHAP-by-category analysis (Fig. 21). For the model's strongest feature `thal`, the median SHAP

Figure 10. Median performance across all metrics for each algorithm under tuned rnCV. LDA achieves the joint-highest AUC and PR-AUC.

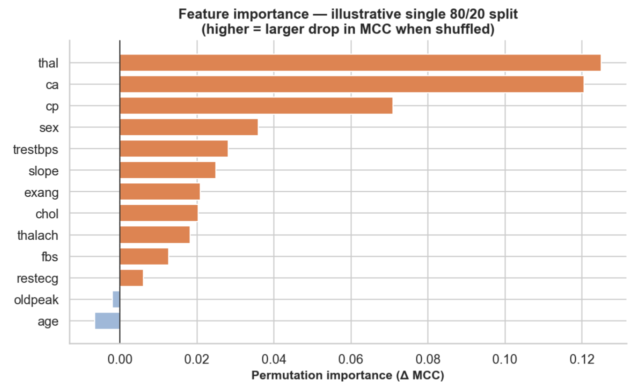


Figure 11. Illustrative permutation importance on a single 80/20 stratified split. Used here to motivate the choice of method only; actual feature selection was performed independently inside each of the 50 outer folds of the rnCV.

value is -0.20 for both TN *and* FN patients, and $+0.28$ for both TP *and* FP patients. In other words, the model's most influential feature pushed FN patients in exactly the same direction as TN patients (toward predicting healthy), and pushed FP patients in exactly the same direction as TP patients (toward predicting disease). The errors occur because `thal` was *misleading* for those specific patients — they are diseased patients whose thallium scan was normal, or healthy patients whose thallium scan was abnormal. This is a textbook clinical reality: thallium scans are imperfect proxies for coronary disease, and the model's reliance on this strong but imperfect feature is the dominant source of its errors.

The same pattern is visible at the level of raw clinical-feature values (Fig. 22), confirming that the SHAP findings reflect a real clinical signal rather than an artefact of how the model uses the features. FN patients exhibit `thal` $= 3$ (normal scan) and low `ca` values (median ~ 1 vessel), almost indistinguishable from TN patients — yet they were truly diseased. FP patients exhibit `thal` $= 7$ (reversible defect) and `cp` $= 3$–4 (non-anginal or asymptomatic), patterns that closely resemble TP patients — yet they were healthy. This double confirmation, at both the SHAP-contribution level and the raw-feature level, makes a clear point: the model is not malfunctioning; it is correctly extrapolating from feature values that happen to be misleading for the specific patient. This is the inherent imperfection of any decision-support model that relies on diagnostic findings which are themselves imperfect proxies for the underlying disease.

Figure 23 confirms that the errors cluster near (but not exactly at) the decision boundary: the FN group has predicted probabilities centred around 0.22 with maximum 0.33, while the FP group has probabilities centred around 0.74 with minimum 0.58. None of the errors occurred at very high or very low probabilities, suggesting a probability-based triage strategy may be useful: patients with predicted probability between 0.30 and 0.70 should perhaps be routed to additional clinical review rather than acted on by the model alone.

Clinician-oriented summary. The model bases its decisions on five clinical findings drawn from routine cardiology workup: thallium scan, number of major vessels affected on fluoroscopy, chest pain type, exercise-induced angina, and patient sex. Each contributes in a clinically expected direction. The model is highly reliable when these findings concord (TP and TN groups have well-separated predicted probabilities). Errors occur when the dominant feature, the thallium scan, is misleading for an individual patient: a normal thallium result in a diseased patient, or an abnormal result in a healthy one. Patients with predicted probabilities in the moderate range (0.30–0.70) are those the model itself indicates as ambiguous, and should be subject to additional clinical review. Errors in the present validation set were concentrated in the 50-60 age band and absent in younger and older patients. A bias toward false negatives over

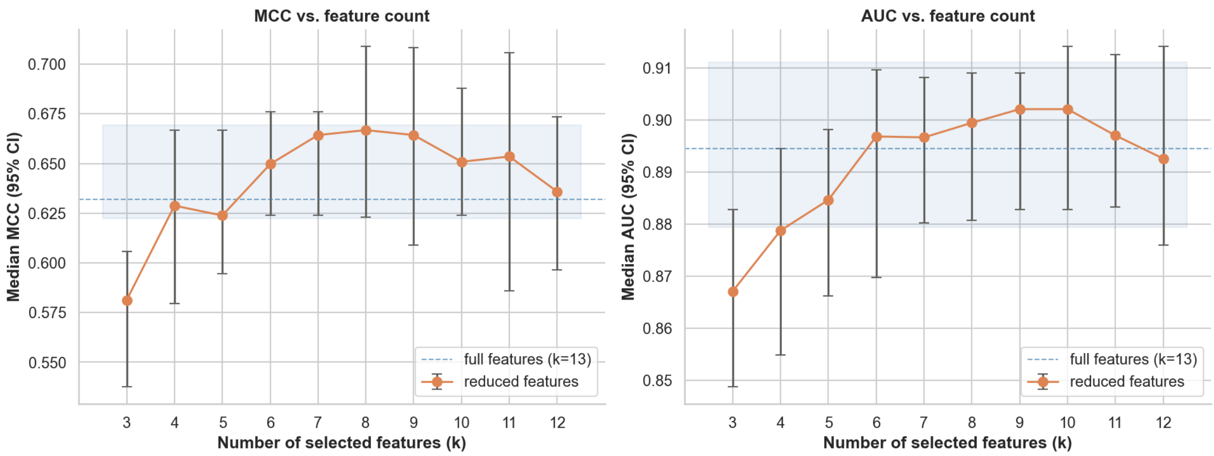


Figure 12. Performance versus number of selected features k. Median MCC peaks at $k = 8$; AUC peaks at $k = 9$. The full-feature ($k = 13$) baseline is shown as a reference band.

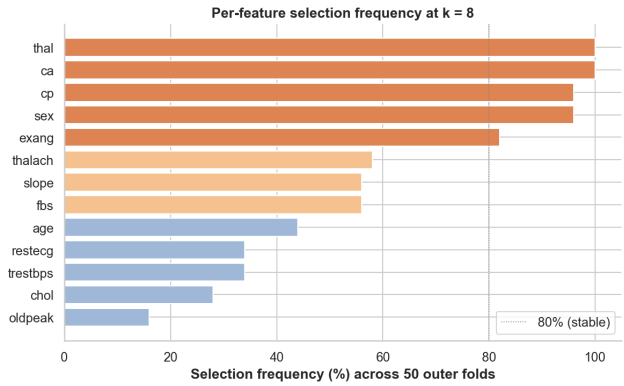


Figure 13. Per-feature selection frequency at $k = 8$ across all 50 outer folds (10 repetitions × 5 outer folds). Five features clear the 80% stability threshold and constitute the stable feature set used for deployment.

false positives (6:4 ratio) suggests that any deployment should consider lowering the decision threshold below 0.5 to favour sensitivity. The model is suitable as a clinical decision-support aid but should not be used as a stand-alone diagnostic tool.

4 Conclusions

4.1 Summary of Main Findings

This work developed and rigorously evaluated a heart disease classification pipeline on the Cleveland subset of the UCI Heart Disease dataset. The main findings are:

1. Among seven candidate algorithms evaluated under repeated nested cross-validation, Linear Discriminant Analysis with shrinkage was identified as the winning algorithm, achieving median MCC 0.632 [0.623, 0.669], AUC 0.895 [0.879, 0.911], and PR-AUC 0.894 [0.875, 0.916] across 50 outer-test scores. The top six algorithms had overlapping 95% confidence intervals on median MCC, but LDA achieved the joint-highest AUC and PR-AUC. The result confirms that on small clinical tabular datasets, simple linear methods can match or exceed gradient-boosting ensembles — the strong distributional assumptions of LDA act as a useful regulariser when sample size is limited.

2. Permutation-importance feature selection embedded in the outer loop identified a stable feature set of five categorical variables — `thal`, `ca`, `cp`, `sex`, and `exang` — selected in $\geq 80\%$ of the 50 outer folds at the optimal $k = 8$. Reducing to this stable feature subset slightly improved median MCC ($0.632 \rightarrow 0.667$) over the full-feature baseline. Notably, none of the continuous features entered the stable set, indicating that the categorical findings of routine cardiology workup carry sufficient information for reliable classification on this dataset.

3. Three independent analyses — EDA-based statistical association, model-driven feature selection inside rnCV folds, and post-hoc SHAP interpretation on the final model — converged on the same five features as the most informative predictors. This convergence strengthens

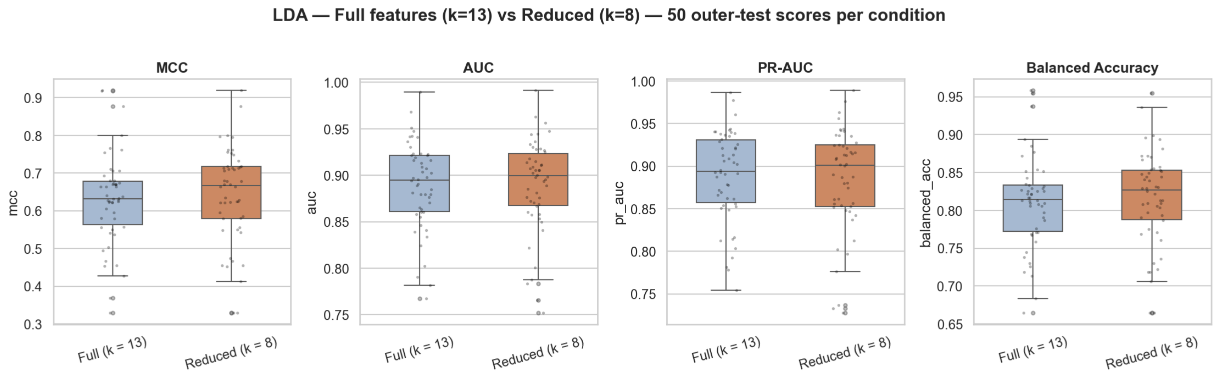


Figure 14. LDA performance on the 13-feature full set versus the $k = 8$ reduced set across 50 outer-test scores. Median MCC improves slightly with reduction ($0.632 \to 0.667$), with all other metrics either improving or holding constant (Table 4).

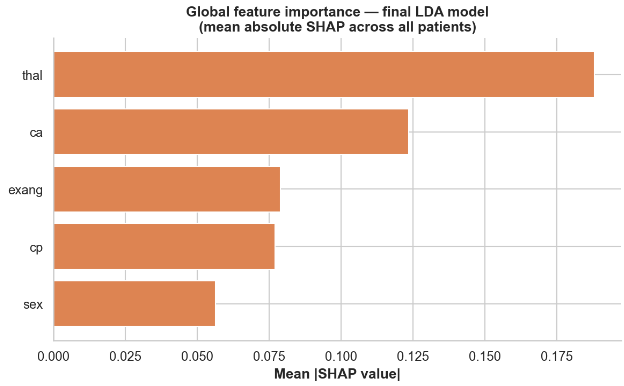


Figure 15. Global feature importance from SHAP values, computed for the final model on all 242 patients. The thallium scan dominates, followed by vessel count and exercise-induced angina.

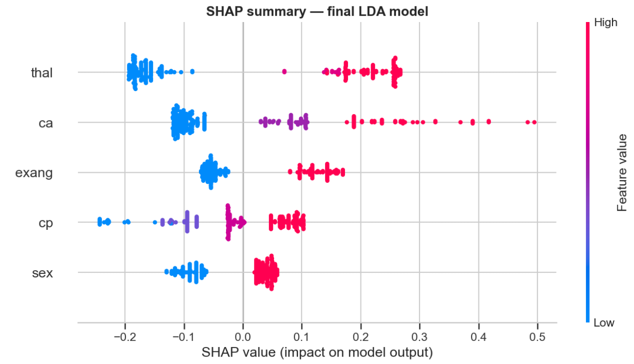


Figure 16. SHAP summary (beeswarm) plot for the final model. Each point is one patient; horizontal position is the SHAP value; colour encodes feature value (red = high, blue = low). All five features push predictions in clinically expected directions.

confidence that the model exploits genuine clinical signal rather than dataset artefacts.

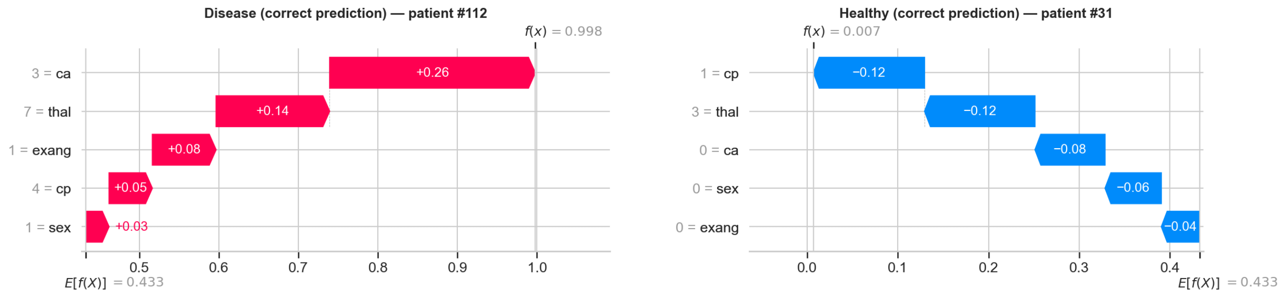
4. The SHAP analysis revealed that `cp` contributes to predictions in the clinically expected direction, with asymptomatic chest pain (code 4) pushing predictions toward disease and typical angina (code 1) pushing away. This counter-intuitive but well-documented *silent ischaemia* pattern is consistent with the cardiology literature and the selection bias of a clinical referral cohort.

5. The bonus error analysis on a held-out 30% validation split achieved 86.3% accuracy with a 6:4 false-negative-to-false-positive ratio. The dominant error mechanism was identified through SHAP-by-category analysis: misclassifications occur when the dominant feature (thallium scan) is misleading for an individual patient. Errors clustered near the decision boundary (predicted probability 0.30–0.70) and in the 50-60 age band, suggesting a probability-based triage strategy and additional caution in this demographic subgroup.

6. The deployable pipeline is a complete sklearn `Pipeline` that accepts raw 13-column input, internally restricts to the stable feature subset, performs imputation and scaling, and produces probabilistic predictions. Reload-and-predict verification on raw input confirmed deployability. This addresses the most common failure mode of saved-model artefacts in machine-learning practice.

4.2 Limitations

Several limitations should be acknowledged. The dataset comprises only 242 patients from a single



(a) Correctly predicted disease patient (#112), $P = 0.998$. **(b)** Correctly predicted healthy patient (#31), $P = 0.007$.

Figure 17. Individual SHAP waterfall plots. Each bar shows how a single feature value moves the prediction up or down from the model's baseline.

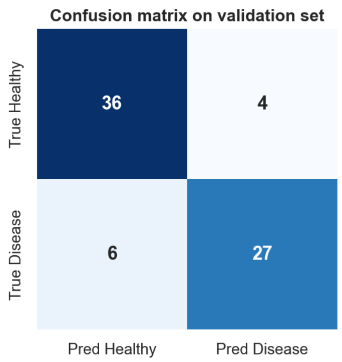


Figure 18. Confusion matrix on the held-out validation set. False negatives outnumber false positives 6:4.

institution, all of whom were referred for invasive angiography. Generalisation to populations with different baseline risk (e.g., primary care screening) or different demographic compositions is unverified. The rnCV procedure provides an unbiased generalisation estimate *within* the dataset's distribution but cannot establish external validity — this would require independent validation on at least one additional cohort.

The 80% stability-selection threshold, while supported by a clear bimodal gap in the selection-frequency distribution at $k = 8$, remains a methodological choice. A sensitivity analysis varying the threshold across $\{70, 80, 90\}\%$ would strengthen the case that the stable feature set is robust to this choice.

LDA assumes Gaussian-distributed class-conditional features with shared covariance structure — assumptions that are only approximately satisfied here. Although LDA outperformed more flexible alternatives in cross-validation, model calibration was not formally assessed. Predicted probabilities should be interpreted with appropriate caution before use in any decision-theoretic clinical workflow that combines them with prior risk estimates.

SHAP values are local linearisations of the model's behaviour around individual predictions; they describe the model, not the underlying biology. Strong feature contributions in the SHAP analysis should not be interpreted as causal effects on disease pathogenesis. The KernelExplainer additionally depends on the choice of background distribution — here summarised by 50 k-means centroids — and SHAP values may shift modestly under different background choices.

Finally, the bonus error analysis was conducted on a single 70/30 split with only $\sim$10 misclassifications. The patterns observed (FN / FP ratio, age-band concentration, thallium-scan error mechanism) should be interpreted as hypothesis-generating signals rather than definitive findings. Larger validation cohorts would be required to confirm them.

4.3 Lessons Learned

Several methodological lessons emerged from this work. First, the discipline of placing every preprocessing and feature-selection step *inside* the cross-validation pipeline prevents the form of optimistic bias that frequently inflates published performance estimates. Even minor leakage — such as fitting an imputer on the full development set before splitting — can shift reported MCC by clinically meaningful amounts.

Second, hyperparameter tuning helps tree-based methods more than linear ones. The fact that

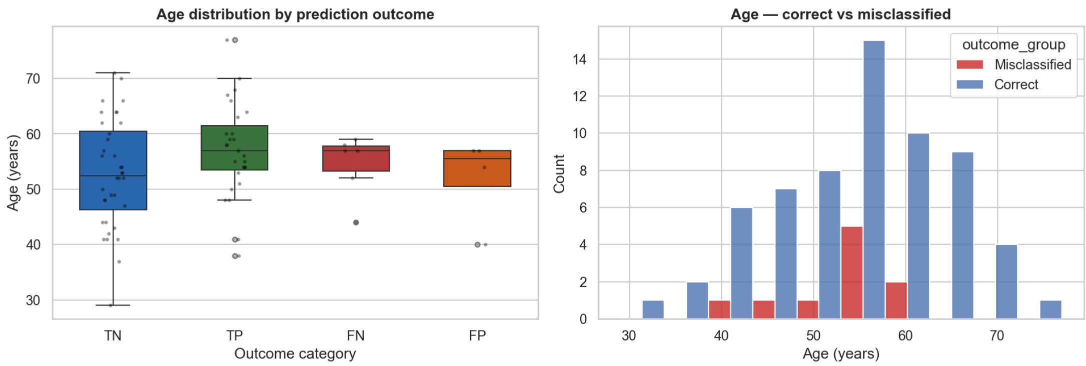


Figure 19. Age distribution by error category (left) and the histogram of correct versus misclassified patients (right). Errors cluster in the 50-60 age band.

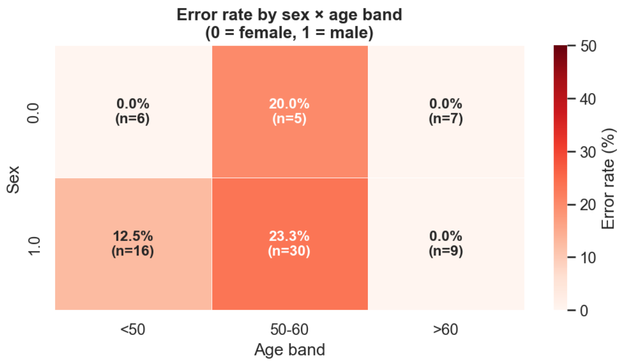


Figure 20. Error rate stratified by sex × age band on the validation set. The 50-60 age band shows the highest error rate (∼20%) for both sexes; both younger and older patients are classified essentially without error.

LDA and GaussianNB performed identically at default and tuned hyperparameters reflects their small effective hyperparameter space; the dramatic gains seen for XGBoost and CatBoost reflect the opposite.

Third, the most useful interpretability evidence is convergence across methods. Any single method — EDA correlations, permutation importance, or SHAP values — can produce noisy or misleading rankings. The agreement between three independent methods on the same five features was the strongest argument that the model captures real clinical signal.

4.4 Suggested Further Research

Several directions for further work are suggested by this study:

- **Threshold optimisation** based on a clinical utility function explicitly weighting false negatives more heavily than false positives, rather than the default 0.5 threshold.
- **Subgroup-specific performance assessment** on larger samples to formally evaluate whether the 50-60 age-band error concentration observed in the bonus analysis is genuine.
- **Causal analysis** using observational causal inference techniques to investigate whether the model's strong reliance on thallium scan reflects a causal effect or a downstream marker of underlying disease.

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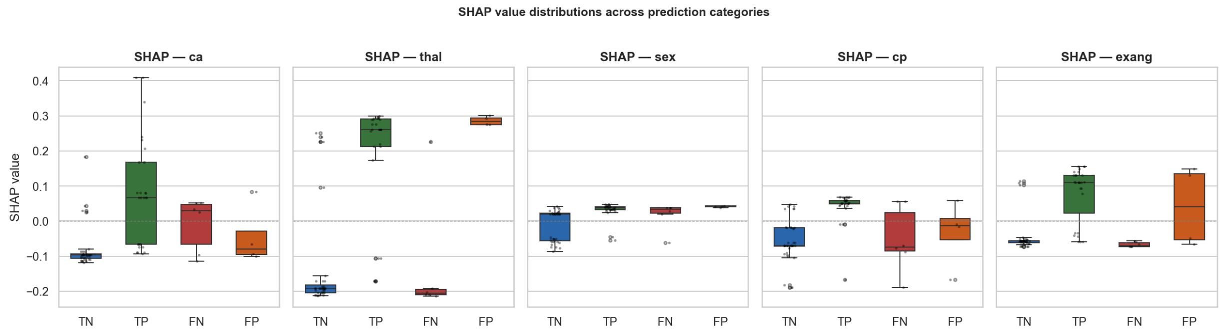

Figure 21. SHAP value distributions across the four error categories (TN, TP, FN, FP). The `thal` panel reveals the dominant error mechanism: FN patients have SHAP values nearly identical to TN patients, indicating that the thallium scan was the primary driver of the misclassification.

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AI Usage Disclosure

This work was developed with the assistance of an AI tool (Anthropic Claude) throughout the project. The contributions of the AI tool and the author were as follows.

AI-assisted contributions. The AI tool was used to: (i) draft the skeletal structure of the OOP code (ii) draft the figure generation code, (iii) discuss design trade-offs (e.g. choice of inner-loop metric, scaler choice, feature-selection criterion, k-selection criterion).

Author contributions. The author was responsible for: (i) all final design decisions (algorithm choices, hyperparameter ranges, criteria for k-selection, the 80% stability threshold, model deployment design), (ii) running all notebooks on the actual dataset, generating all numerical results and figures, (iii) verifying outputs and catching errors (including identification of an inconsistency in saved feature-selection metadata), (iv) clinical interpretation of results in light of cardiology knowledge (e.g. the silent-ischaemia interpretation of the chest-pain feature).

Methodology of use. The AI tool was used as a thinking partner: design choices were dis-

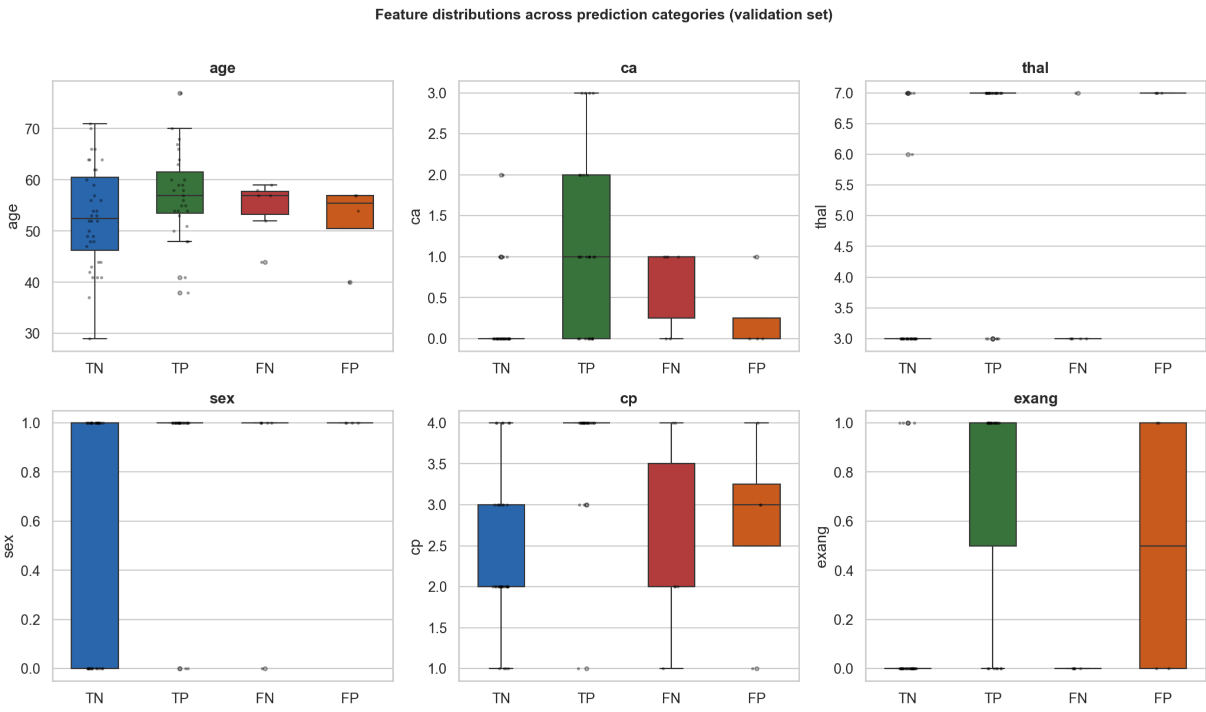

Figure 22. Raw clinical-feature distributions across the four error categories (TN, TP, FN, FP) on the validation set. FN patients (red) closely resemble TN patients (blue) on the dominant features `thal` and `ca` despite being truly diseased; FP patients (orange) closely resemble TP patients (green) on the same features despite being healthy. This explains the SHAP-by-category pattern (Fig. 21) at the level of underlying clinical findings.

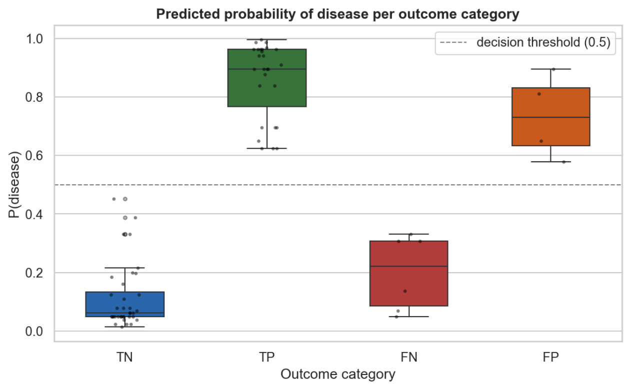

Figure 23. Predicted probability of disease per outcome category. Errors occupy the moderate-probability region (FN: 0.05–0.33; FP: 0.58–0.89), suggesting a clinical triage strategy.

cussed before being implemented, and the author actively pushed back on suggestions that did not fit the assignment requirements or established clinical context. Code generated by the tool was always reviewed and verified by running on the actual data; results were never accepted without confirmation. No section of this report was generated without active editorial input from the author.

Tools used. Anthropic Claude (web-based chat interface)